Lifestyle Dataset Analysis

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AMS 597

Introduction

In this project we aim to answers important questions about how Lifestyle factors affect different physiological markers. Using statistical methods such as Linear Regression, Analysis of Variance, and Neural Networks, we aim to derive connections between health and lifestyle. This can be applied in the real world to help create better lifestyle choices by giving answers to our 3 research questions:

Research Question 1

- Predictive Modeling of Energy Expenditure: Can we accurately predict the total calories burned during a workout based on physiological intensity and exercise type? We utilize Bidirectional Stepwise Regression (BIC) to identify the most significant predictors while addressing multicollinearity.

Research Question 2

- Inference of Physiological Differences: Does resting heart rate vary significantly across different age groups and genders? Using an Analysis of Multi-Factor Experiments (ANOVA), we observe the main effects and potential interactions between demographic factors and cardiovascular health.

Research Question 3

- Advanced Non-linear Classification of Body Composition: Is it possible to predict an individual's Body Mass Index (BMI) using only lifestyle and dietary factors, excluding direct physical measurements (no height or weight)? We implement a Neural Network (Multi-Layer Perceptron) with cross-validated hyperparameter tuning to capture non-linear relationships that traditional linear models may miss.

Dataset

This project analyzes a comprehensive, real-world health and fitness tracking dataset consisting of 20,000 observations of 54 variables. It covers a wide range of metrics related to three main categories:

physical characteristics, exercise performance, and nutritional habits. Examples of variables are as follows:

- **Physiological Markers**: Age, Gender, Weight (kg), Height (m), BMI ...
- **Exercise Activity**: Session_Duration, Experience Level, Workout_Type ...
- **Nutrition and Diet**: Diet_type, Carbs, Protein, Fat, Sodium...

Preprocessing

Our dataset is "messy and dirty," requiring significant preprocessing to handle high-cardinality text, multicollinear engineered features, and data leakage. We will drop columns:

- Derived from or containing the target (`Calories_Burned` for Question 1). Including these would give an artificially inflated R^2 and a useless model.

Column	Reason for Drop
Burns_Calories_Bin	Binned version of the target
expected_burn	Formula-based estimate of the target
cal_balance	= Calories_meal − Calories_Burned (contains target)
Burns.Calories..per.30.min._bc	Rescaled version of a near-target column

- Multicollinear features computed directly from other predictors already in the dataset, so keeping them would double-count information and inflate VIF. We remove these features to improve interpretability and reduce redundancy.

Column	Derived from
BMI_calc	Weight / $Height^2$ (same as BMI)
cal_from_macros	Carbs, Proteins, Fats

pct_carbs	Carbs, cal_from_macros
protein_per_kg	Proteins, Weight
pct_HRR	Avg_BPM, Resting_BPM, Max_BPM
pct_maxHR	Avg_BPM, Max_BPM
lean_mass_kg	Weight, Fat_Percentage

- High-cardinality text columns with 36–55 unique levels are not useful to us, dummy-encoding them would add too many columns of noise. The information is already captured by lower-cardinality variables like Workout_Type, Body.Part, and Difficulty.Level. Examples of columns we are dropping:

High-Cardinality Variable
meal_name
Name.of.Exercise
Benefit
Target.Muscle.Group
Equipment.Needed
Workout

Next, we encode categorical variables. For ordinal variables like Difficulty.Level, (Beginner, Intermediate, Advanced) we encode 0,1,2. For binary variables like Gender, we encode 0 or 1.

Lastly, we check for missing data and duplicate rows. None were found.

````
```{r}
cat("Missing values:", sum(is.na(cleaned_data)), "\n")
cat("Duplicate rows:", sum(duplicated(cleaned_data)), "\n")
```

 Missing values: 0
 Duplicate rows: 0
````

Figure 1: Number of Missing Values and Duplicate Rows

Here is our final cleaned dataset (20,000 observations of 37 variables):

```
'data.frame':	20000 obs. of  37 variables:
 $ Age                         : num  34.9 23.4 33.2 38.7 45.1 ...
 $ Gender                      : int  1 0 0 0 1 0 1 0 0 0 ...
 $ Weight..kg.                 : num  65.3 56.4 59 93.8 52.4 ...
 $ Height..m.                  : num  1.62 1.55 1.67 1.7 1.88 1.84 1.78 1.63 1.79 1.6 ...
 $ Max_BPM                     : num  189 179 175 191 194 ...
 $ Avg_BPM                     : num  158 132 124 155 153 ...
 $ Resting_BPM                 : num  69 73.2 55 50.1 70.8 ...
 $ Session_Duration..hours.    : num  1 1.37 0.91 1.1 1.08 0.69 1.67 1.01 1.76 1.17 ...
 $ Calories_Burned             : num  1081 1810 802 1451 1166 ...
 $ Workout_Type                : Factor w/ 4 levels "Cardio","HIIT",..: 3 2 1 2 3 4 3 4 3 3 ...
 $ Fat_Percentage              : num  26.8 27.7 24.3 32.8 17.3 ...
 $ Water_Intake..liters.       : num  1.5 1.9 1.88 2.5 2.91 2.91 2.71 2.88 3.49 2.49 ...
 $ Workout_Frequency..days.week.: num  3.99 4 2.99 3.99 4 3.02 4.96 3.97 4.01 2 ...
 $ Experience_Level            : num  2.01 2.01 1.02 1.99 2 1 3 2.01 3.02 1 ...
 $ BMI                         : num  24.9 23.5 21.1 32.5 14.8 ...
 $ Daily.meals.frequency       : num  2.99 3.01 1.99 3 3 2.99 2.02 2.99 3 3.01 ...
 $ Physical.exercise           : num  0.01 0.97 -0.02 0.04 3 -0.04 -0.03 0 0.02 0.02 ...
 $ Carbs                       : num  268 214 246 203 333 ...
 $ Proteins                    : num  106 85.4 98.1 80.8 133.1 ...
 $ Fats                        : num  71.6 57 65.5 54.6 88.4 ...
 $ Calories                    : num  1806 1577 1608 2657 1470 ...
 $ meal_type                   : Factor w/ 4 levels "Breakfast","Dinner",..: 3 3 1 3 1 4 1 4 3 3 ...
 $ diet_type                   : Factor w/ 6 levels "Balanced","Keto",..: 5 6 4 4 5 2 3 3 5 4 ...
 $ sugar_g                     : num  31.77 12.34 42.81 9.34 23.78 ...
 $ sodium_mg                   : num  1730 693 2142 123 1935 ...
 $ cholesterol_mg              : num  285.1 300.6 215.4 9.7 116.9 ...
 $ serving_size_g              : num  120.5 109.2 399.4 314.3 99.2 ...
 $ cooking_method              : Factor w/ 7 levels "Baked","Boiled",..: 4 3 2 3 1 7 3 5 2 4 ...
 $ prep_time_min               : num  16.2 16.5 54.4 27.7 34.2 ...
 $ cook_time_min               : num  110.79 12.01 6.09 103.72 46.55 ...
 $ rating                      : num  1.31 1.92 4.7 4.85 3.07 3.38 3.81 3.16 2.81 1.6 ...
 $ Sets                        : num  4.99 4.01 5 4.01 4.99 4 5.01 4.97 3.99 4.02 ...
 $ Reps                        : num  20.9 16.1 21.9 16.9 15 ...
 $ Burns.Calories..per.30.min. : num  343 357 360 352 329 ...
 $ Difficulty.Level            : int  2 1 1 2 2 0 2 1 0 1 ...
 $ Body.Part                   : Factor w/ 7 levels "Abs","Arms","Back",..: 6 4 2 7 1 2 7 3 4 5 ...
 $ Type.of.Muscle              : Factor w/ 13 levels "Anterior","Grip Strength",..: 4 4 2 11 13 13 13 5 7 6 ...
```

Figure 2: Structure of Cleaned Lifestyle Dataset

Before answering each research question, we perform Exploratory Data Analysis (EDA) related to our question. For the entire dataset, we create a Correlation Plot to observe how each variable is correlated to each other:

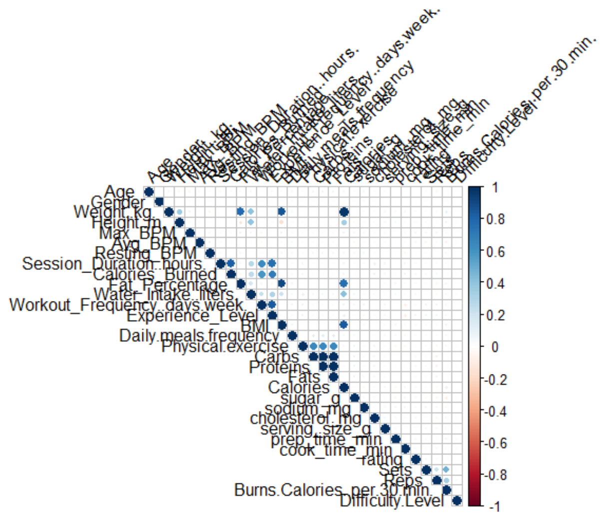


Figure 3: Correlation Plot of Lifestyle Data

Research Question 1: *Can we accurately predict the total calories burned during a workout based on physiological intensity and exercise type?*

EDA

For examination of the data, it would be useful to explore what variables could have the most impact on the Calories_Burned variable. To do this, we can rank each variable by their correlation coefficient with Calories_Burned with a correlation plot:

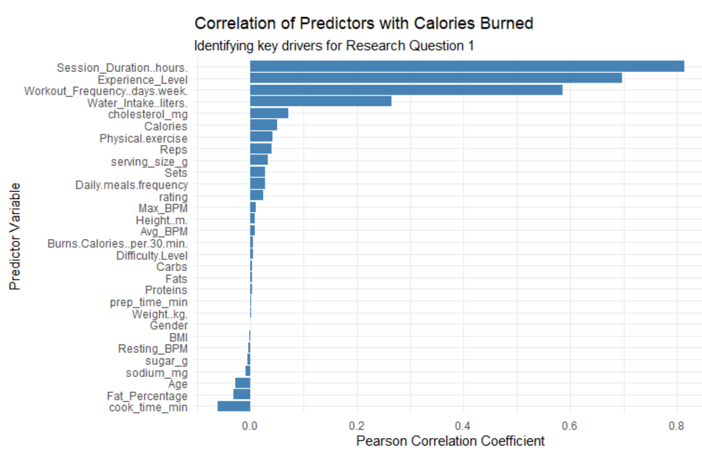


Figure 4: Correlation of Predictors with Calories Burned

We observe that Session_Duration..hours, Experience_Level, Workout_Frequency..days.week, and Water_Intake..liters are the most significant variables to predict Calories_Burned. We would expect these variables to be selected for our linear regression model.

We hypothesized Workout_Type would significantly affect Calories_Burned. To visualize this we created a density plot measuring the density of Calories_Burned by Workout_Type.

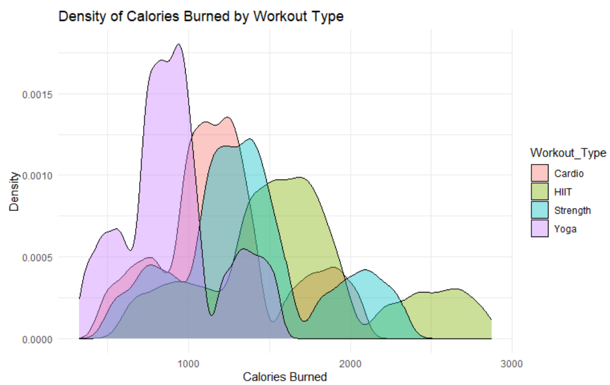


Figure 5: Density of Calories Burned Based on Workout_Type

The density plot shows us that workouts such as yoga do not burn as many calories while HIIT and Strength have higher calories burned. This suggests that Workout_Type may be one of our key predictors.

Methodology

Data Partitioning and Scaling. The cleaned dataset was partitioned into training (80%) and test (20%) sets using stratified random sampling on the response variable, Calories_Burned, preserving the shape of the outcome distribution in both splits. All numeric predictors of the training set were then exclusively standardized using Z-score normalization, to ensure that no test-set information leaks into the preprocessing pipeline.

Full Model as Baseline. An ordinary least squares (OLS) regression was first estimated using all available predictors after preprocessing. This full model serves as an upper-bound reference against which the variable-selected model can be compared in terms of adjusted R^2 and AIC.

Variable Selection: Bidirectional Stepwise Regression with BIC. The BIC criterion ($k = \log n$, where n is the number of training observations) was chosen over AIC because BIC imposes a substantially heavier penalty per additional parameter at large sample sizes, steering the selection toward more parsimonious models. The stepAIC function from the MASS package was run in bidirectional mode, meaning predictors could be added to or removed from the current model at each step depending on which action most reduces BIC. Bidirectional search is preferred over purely forward or purely backward selection because it can recover from early inclusion or exclusion mistakes: a predictor that appeared unhelpful alongside a strongly correlated variable may become useful once that variable is dropped, and vice versa. The procedure terminates when no single addition or removal improves BIC.

Multicollinearity Diagnostics. Following variable selection, the Variance Inflation Factor (VIF) was computed for each retained predictor using the vif function from the car package. For factor variables with multiple degrees of freedom, the Generalized VIF was used. A VIF threshold of 10 was applied as the criterion for serious multicollinearity; values between 5 and 10 were flagged as moderate concerns.

Test Set Evaluation. Predictions from the final selected model were generated on the held-out test set in standardized units and then back-transformed to the original calorie scale using the training mean and standard deviation of Calories_Burned. Performance was quantified using three metrics: Root Mean Square Error (RMSE, in calories), Mean Absolute Error (MAE, in calories), and the coefficient of determination (R^2).

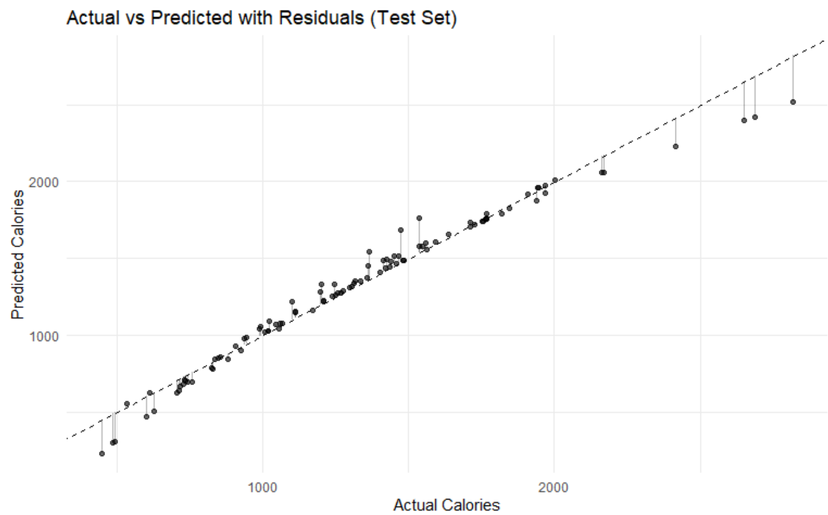


Figure 6: Actual vs Predicted calories plot

```
=== FINAL MODEL SUMMARY ===
Method         : Bidirectional Stepwise (BIC)
Predictors     : 9
Test R²        : 0.9684
Test RMSE      : 90.05 calories
```

Figure 7: Final Model Summary Bidirectional Stepwise (BIC) Regression

The final model achieved a test R^2 of **0.9684**, indicating that approximately **96%** of the variance in calories is explained by the selected variables. The test RMSE was 90.05 calories, and the MAE was **58.93** calories. The actual vs predicted plot shows that predictors cluster tightly around the 45-degree reference line across the full range of the response, with no systematic bias.

The dominance of session duration, workout frequency, and experience level as predictors is physiologically coherent: longer and more frequent sessions accumulate more energy expenditure. In conclusion, the results indicate that calorie expenditure during exercise can be predicted with strong accuracy from a parsimonious set of physiological intensity and behavioral variables, without requiring any features prone to data leakage.

Research Question 2: Does *resting heart rate vary significantly across different age groups and genders?*

To examine how Resting Heart Rate (BPM) and BMI vary across different age groups, we use violin plots and histograms to visualize their distributions. These plots allow us to assess not only central tendencies (such as mean and median) but also the spread, density, and shape of the data within each group.

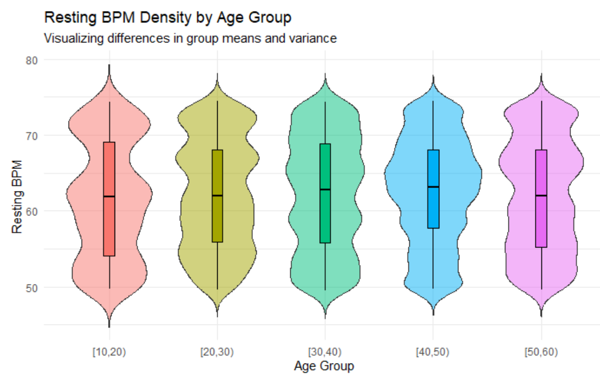


Figure 8: Violin Plot of Resting BPM by Age Group

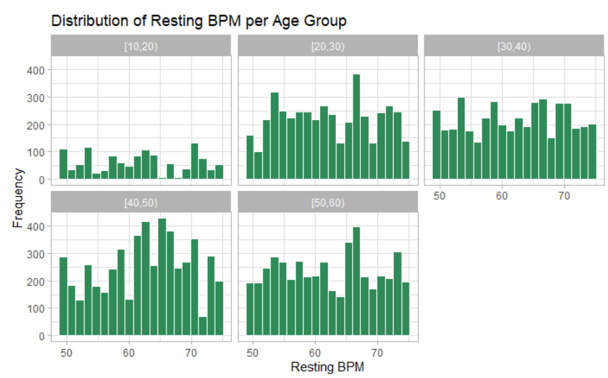


Figure 9: Histograms displaying distributions of Resting BPM for each age group

Overall, while the central tendencies (means/medians) of BMI across age groups are similar, the distributional shapes differ slightly, particularly in terms of density. This suggests that age may not strongly affect the average BMI, but it could influence the variability and distribution pattern within groups.

Methodology

To answer this question, it was decided to perform a multi-factor ANOVA test to determine if age and gender affect resting BPM. Before the test can be conducted, age needs to be turned into a categorical variable. To do this, an additional column was added to the dataset, placing each person into a 10-year age group depending on the age given.

Age	Age_group
34.91	[30,40)
23.37	[20,30)
33.20	[30,40)
38.69	[30,40)
45.09	[40,50)
53.19	[50,60)
23.17	[20,30)
55.92	[50,60)
24.11	[20,30)
39.19	[30,40)
50.51	[50,60)
47.77	[40,50)
19.00	[10,20)
38.01	[30,40)

Figure 10: A person's age and their corresponding age group.

Now that we have a categorical variable for someone's age, the ANOVA test can be done. An alpha level of 0.05 is used for determining the results.

```
                 Df  Sum Sq Mean Sq F value   Pr(>F)
Age_group:Gender   4     309    77.2   1.455    0.213

H0: There are no interactions between gender and age groups.
Ha: There are interactions between gender and age groups.
```

Figure: 11: Model results for age group and gender determining resting bpm.

Since p > 0.05 we do not reject H0. There is no statistical significance that there is any interaction between age group and gender when comparing the means of average bpm.

```
                 Df  Sum Sq Mean Sq F value   Pr(>F)
Gender            1      10     9.6   0.181    0.670

H0: The means between genders are equal
Ha: The means between genders differ.
```

Figure 12: Model results for gender determining resting bpm.

For the main effect of Gender, p > 0.05 so we do not reject H0. There is no statistical significance that gender affects the mean response of resting bpm.

```
                    Df  Sum Sq Mean Sq F value   Pr(>F)
Age_group            4    1279   319.7   6.023 7.71e-05 ***

H0: The means between age groups are equal.
Ha: At least one mean differs between the age groups.
```

Figure 13: Model results for age group determining resting bpm.

For the main effect of Age group, $p < 0.05$ so we reject H0. This means there is statistical significance that the age group you belong in effects the mean response of resting bpm.

Results

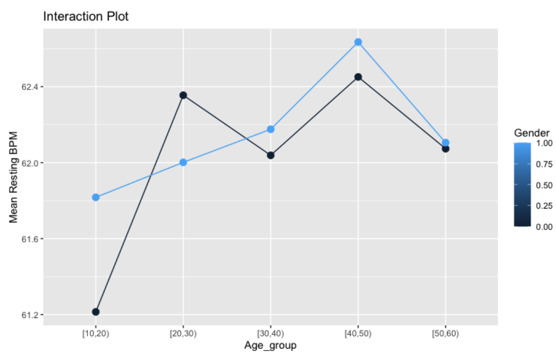


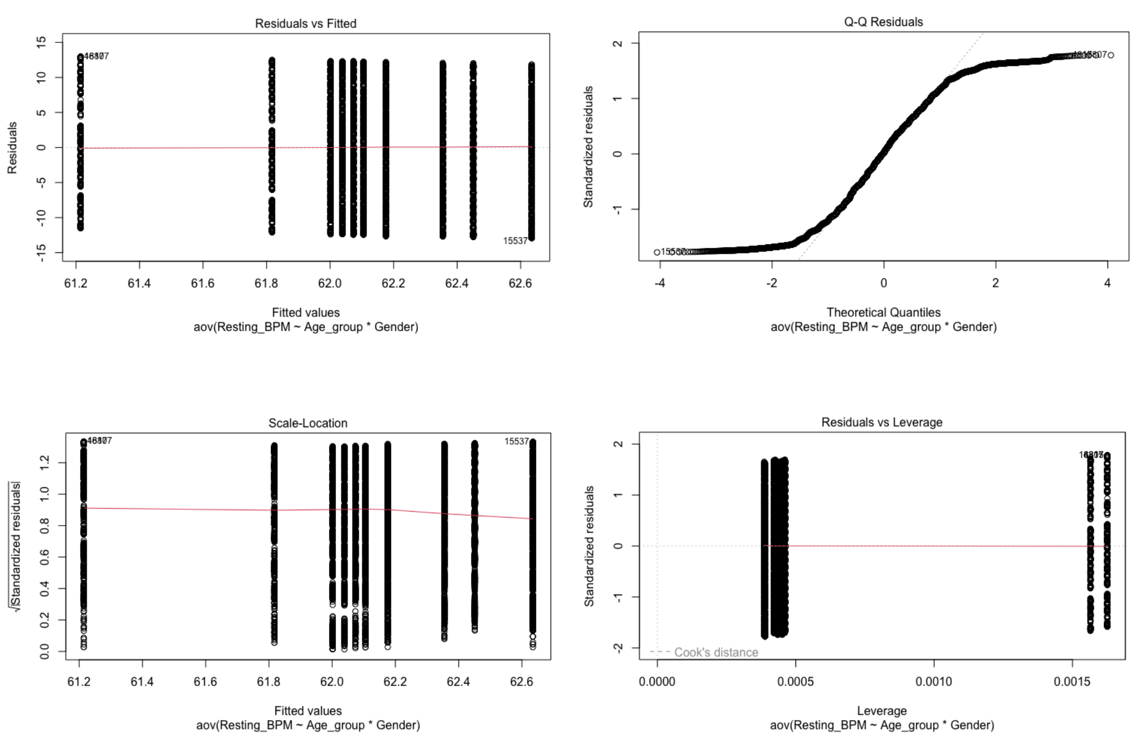
Figure 14: Interaction plot for interaction of age group and gender on resting bpm.

Since the ANOVA test failed to show any significance of age group and gender affecting resting bpm, we will look at the distances between the genders and their corresponding age group. The biggest difference occurs between male and females in the age range of [10,20). I believe this occurs because the data only contains ages from 18 to 59. As a result, the age group [10,20) has the smallest sample size out of all the other age groups. Other than the age group mentioned, the rest of the distances between gender seem to be relatively small and consistent across age groups.

```
$Age_group
                        diff         lwr         upr     p adj
[20,30)-[10,20)  0.66159906  0.02651082  1.29668731 0.0362941
[30,40)-[10,20)  0.58621458 -0.05021966  1.22264881 0.0878002
[40,50)-[10,20)  1.02084715  0.39513548  1.64655882 0.0000841
[50,60)-[10,20)  0.56768395 -0.06368367  1.19905158 0.1016634
[30,40)-[20,30) -0.07538448 -0.49772788  0.34695891 0.9885859
[40,50)-[20,30)  0.35924809 -0.04675746  0.76525363 0.1115794
[50,60)-[20,30) -0.09391511 -0.50858424  0.32075402 0.9723014
[40,50)-[30,40)  0.43463257  0.02652479  0.84274035 0.0301618
[50,60)-[30,40) -0.01853062 -0.43525829  0.39819704 0.9999513
[50,60)-[40,50) -0.45316320 -0.85332380 -0.05300259 0.0171700
```

Figure 15: Tukey test for age groups at a 95% CI.

When looking at the Tukey test for age groups, many of the confidence intervals contain 0. This means that there are differences between group means. However, every single CI that contains 0 also has a p-value > 0.05 which means that this result is not statistically significant. All the confidence intervals without 0 (significant differences between the two groups) have a p value < 0.05 meaning they are statistically significant. These statistically significant values are [20,30)-[10,20), [40,50)-[10,20), [40,50)-[30,40), and [50,60)-[40,50).



Figures 16-19: (from left to right) Residuals vs Fitted, Q-Q plot, Scale- Location, and Residuals vs Fitted plot.

To check for the assumptions needed for ANOVA to be reliable, four plots have been made. The red line for the Residuals vs Fitted plot is horizontal, so the linearity assumption is met. For the Q-Q plot, most of the data falls along the dotted line; however, the tails do deviate heavily. This means that kurtosis is higher than normal. For the standardized residuals, the red line is mostly horizontal, which means there is an equal variance of residuals. For the leverage plot none of the points fall outside of cook's distance, meaning none of the points are way more influential than the others.

All the assumptions have been met with the slight downside of the data being distributed normally just with higher kurtosis. A solution to this problem could be to log transform the resting bpm to make the tails longer. Overall, the model shows no evidence of any interaction between gender and age groups. There is evidence to support age group effecting resting bpm, but no evidence to show gender effecting this.

Research Question 3: Is it possible to predict an individual's Body Mass Index (BMI) using only lifestyle and dietary factors, excluding direct physical measurements (no height or weight)?

EDA

To explore whether BMI could be predicted from lifestyle-related variables, we first conducted exploratory data analysis on the training data. The purpose of this step was to examine patterns between BMI and several candidate predictors, identify possible nonlinear relationships, and assess whether a flexible machine learning model would be appropriate

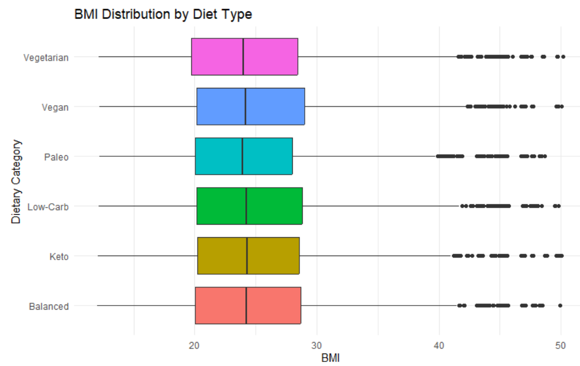


Figure 20: BMI Distributions by Diet Type

We also compared BMI across diet types using a boxplot. While some variation in BMI distributions appeared across dietary categories, the differences were not sharply separated visually. This suggests that diet type alone may not strongly distinguish BMI, but it may still contribute useful information when combined with other predictors in a multivariable model.

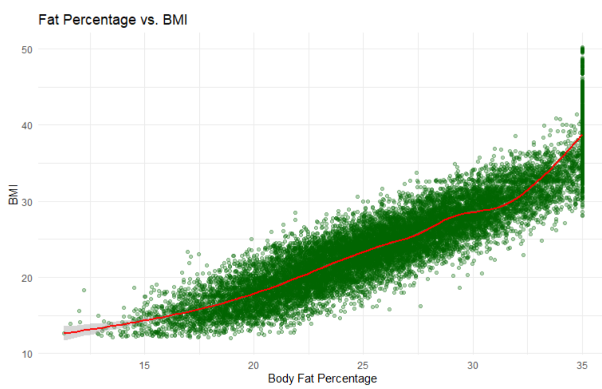


Figure 21: Scatterplot of Fat Percentage vs BMI with GAM curve

Next, we examined the relationship between Fat Percentage and BMI using a scatterplot with a smooth GAM curve. This plot showed a clear positive association, and the smoothed trend suggested that the relationship may not be perfectly linear across the full range of values. This observation supported the use of a neural network, since neural networks are able to capture nonlinear patterns more flexibly than standard linear models.

Methodology

To address Question 3, we aim to use the Neural Network regression model to predict BMI from lifestyle-related and health-related predictors. The response variable was **BMI**, while predictor variables were drawn from the available demographic, dietary, exercise, and body-composition features in the dataset. To avoid direct data leakage and possible hidden calculation, **height** and **weight** were excluded from the model because BMI is directly derived from these measurements. We later discovered that **Calories** should be also removed because it appeared to act as a strong proxy for BMI and produced suspiciously high predictive performance (0.956) when included. To perform this model, each categorical variable needs to be converted into dummy variables. All predictors were then **centered and scaled** based on the training data, since neural networks generally perform better when inputs are on comparable numerical scales. The data was separated into **training** and **test** sets so that final performance could be evaluated on unseen observations. Packages such as "caret" framework and "nnet" method were used to implement the mode since BMI is a continuous outcome measure. Model tuning was performed using **5-fold cross-validation** over different combinations of **hidden units** and **weight decay**. To reduce computational burden, model training was conducted on a random subset of the training data, while final evaluation was still performed on the held-out test set. Model performance was assessed using **R^2**, **RMSE**, and **MAE**. In addition, V**ariable Importance** plot was made to identify which predictors have the strongest importance to BMI prediction

Results

```
=== FINAL MODEL SUMMARY ===
Method            : Neural Network
Test R²           : 0.814
Test RMSE          : 2.93 BMI units
Test MAE           : 2.27 BMI units
```

Figure 22: Final Model Summary

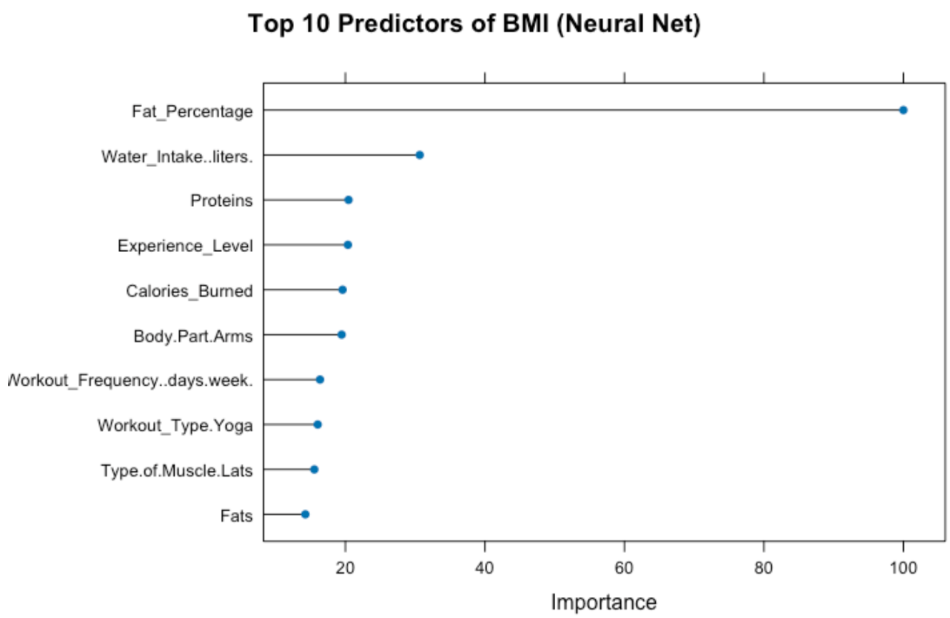


Figure 23: BMI Predictors Importance Order

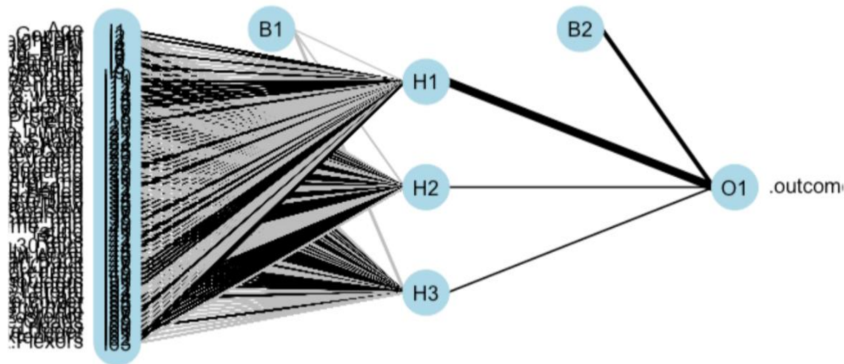


Figure 24: Neural network architecture for BMI prediction

The neural network model was tuned using 5-fold cross-validation over different combinations of hidden units and weight decay. The final model was selected based on the lowest cross-validated RMSE. By looking at Figure 22, the neural network achieved an **R^2 of 0.814**, indicating that the model explained about **81.4% of the variation in BMI**. The **RMSE was 2.93**, and the **MAE was 2.27**, meaning that the predicted BMI values were typically within about 2 to 3 BMI units of the observed values. These results suggest that the neural network provided reasonably strong predictive performance for BMI. Figure 23 showed that **Fat_Percentage** was the most influential predictor of BMI in the neural network model, with a substantially higher importance score than the remaining variables. Other predictors, including **Water_Intake**, **Proteins**, **Experience_Level**, and **Calories_Burned**, also contributed to prediction, but their relative importance was much smaller. This suggests that while BMI prediction benefited from multiple predictors, **fat percentage carried the strongest predictive signal** in the model.

For figure 24, it displays the input layer, hidden layer, and output node of the fitted neural network. Because the predictors were dummy encoded, the input layer contains many nodes, making the architecture difficult to interpret visually.

Limitations

- **High-Cardinality Variable Removal:** Some variables were dropped because they contained too many unique levels. This means specific nuances of individual exercises or meals were lost to favor simplicity
- **Categorical Simplification:** Ordinal variables like Difficulty.Level were simplified into numeric codes (0, 1, 2). This assumes the "distance" between Beginner and Intermediate is mathematically the same as the distance between Intermediate and Advanced, which may not accurately reflect physiological reality
- Each methodology had its own inherent limitations:
    - **Linear Assumption (OLS):** Assumed linear relationships between variables such as Session_Duration and Calories_Burned which may not accurately reflect the real relationship. The stepwise variable selection is sensitive to small changes in the dataset and may select different predictors if the data were resampled. Even though we addressed multicollinearity, it can still exist which can affect interpretability of regression coefficients.
    - **Neural Network Training Subset:** To reduce computational burden we only trained on 2,000 of the 20,000 observations. This may have reduced the models ability to learn from the full diversity of the dataset. In addition, we originally implemented a 10-fold CV to find the optimal value; instead, we adjusted it tong 5-fold CV because of the

training size, and by reducing the fold number it increases the efficiency which is more preferable to larger dataset like ours at the cost of having higher bias.

  - **Neural Network Blackbox:** The multi-layer perceptron was used to model non-linear relationships. These models are less interpretable than linear models and harder to interpret how specific lifestyle factors affect BMI.

- **Proxy Variables:** Calories had to be removed from the BMI prediction because it acted as a strong proxy which gave a very high R^2 (0.956), potentially masking the influence of other lifestyle factors we intended to study

Conclusion

In this project we gave answers to important questions about how Lifestyle factors affect different physiological markers by implementing statistical methods such as Linear Regression, Analysis of Variance, and Neural Networks. These provided insight into the connections between health and lifestyle.

1. Using Ordinary Least Squares (OLS) regression with BIC bidirectional variable selection, we were able to predict the amount of calories burned in a workout using our lifestyle variables with a high degree of accuracy. Our model selected 9 variables and provided a Test R^2 of 0.9684 and a Test RMSE of 90.05 calories. This could be implemented in the real world for applications such as fitness apps to estimate the amount of calories the user will burn in a given workout.
2. Using a multi-factor ANOVA test we were able to determine if age and gender affect resting BPM. We binned the age variable and performed 3 ANOVA tests and found:
   a. There is no statistical significance that there is any interaction between age group and gender when comparing the means of average bpm
   b. There is no statistical significance that gender affects the mean response of resting bpm

c. There is statistical significance that the age group you belong in effects the mean response of resting bpm

This information can be used as a benchmark when determining whether an individual’s resting BPM falls within a normal range for their age group, helping clinicians and fitness professionals identify potential health risks or cardiovascular issues more accurately.

3. Using a Neural Network with 5-fold cross validation we were able to predict a person's BMI using non-physiological factors of weight and height. We achieved an R^2 of 0.814 and a RMSE of 2.93 BMI units. This information can be used to provide insights into what factors are likely to lead to high BMI and encourage healthier lifestyle choices.

References

Jockeroika. (2024). *Life Style Data* [Data set]. Kaggle. https://www.kaggle.com/datasets/jockeroika/life-style-data